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MASTER'S COURSEWORK
Research Project on the Topic
Uplift Modeling in Recommender Systems

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Annotation

The sphere of recommender systems (RecSys) is highly relevant in today's world as it allows businesses to personalise the experience of each user. Standard approaches like content- or collaborative-based filtering try to identify the most relevant suggestions. However, such recommendations are subject to various biases due to the fact that algorithms overlook the causality effect. Adapting the uplift modeling approach allows to estimate the impact suggestions have on user's actions. The current paper presents an extended research based on the findings of author's bachelor thesis. Through the continued use of PropCare with different types of DLCE algorithms the analysis of frequency information addition to the final prediction is studied in two different settings.

Content Warning

The paper uses phrases (their original or paraphrased variant) directly from the author's bachelor's thesis [\[22\]](#).

Keywords

ML, DL, RecSys, Uplift Modeling

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1 Introduction

Recommender systems (RecSys) have evolved into a crucial element of the contemporary data-driven strategy for improving service personalization. The overall aim of the task is to identify the patterns of user habits and likes. The algorithm subsequently chooses the most suitable user-item combination from the available options and delivers a personalized suggestion. The feedback loop updates the interactions, improving the overall quality of the algorithm’s future suggestions.

The execution typically occurs via either collaborative filtering or content-driven approaches. The earlier idea suggests that a positive correlation exists in interactions among users who share similar characteristics. A highly favored method in this domain is the Matrix Factorization (MF) algorithm [14]. The aforementioned method focuses on forecasting and enhancing the click-through rate by presenting a user with comparable items [29, 31].

Every method mentioned neglects a crucial aspect that correlation does not mean causation. The statement indicates that the action taken with an item (such as clicking, rating, purchasing, etc.) could have occurred even without the suggestion of the algorithm. Consequently, the potential bias introduced [9] tends to diminish the quality of the results obtained. Therefore, a conversation about uplift modeling, which represents a causal method in recommendation systems, is necessary to assess the effect recommendations have on user behavior.

By integrating causality into the process, uplift modeling facilitates the attainment of various potential advantages. Firstly, personalization is improved as the suggestions are now defined by increased user focus. Moreover, the rise in efficiency is highlighted as the computational resources focus on the outcomes with the greatest significance. Ultimately, to assess the causal impact of recommendations, new metrics are implemented to measure the effectiveness of various strategies in this domain.

The research on the topic was performed as a continuation of the work performed under the bachelor’s thesis objectives [22],. The current paper is aimed at extending and verifying the conclusions derived previously in a less synthetic setting. Mainly: (i) addition of frequency information to the final prediction of an uplift model leads to better performance (both causal and standard), (ii) modified algorithm loss with the incorporation of relevance scores leads to even better causal performance.

The paper is structured in the following way. In the following Section (2) the overview of existing methods for uplift modeling is presented with the justification of selected methods. Then in Section (3) the mathematical formulation of the problem is described as well as the modifications proposed with the specificity of formalised goals. After this in Section (4) the description of used data

and conducted experiments is presented. The Section (5) serves as the closing one formalizing the conclusions from the results obtained at the current stage and providing ideas for future development.

2 Literature review

The research remains grounded in causality-aware RecSys, which originated from concepts of uplift modeling applications. The main benefit of a causal effect account is that it reduces the influence of different biases (popularity, noise, etc.) [9]. The primary outcome from it relevant to this paper is that it allows for the assessment of the impact that recommendations have on users.

2.1 Existing approaches

As mentioned in Section (1), one of the most famous methods for RecSys is MF [14]. Still, its simplest form presumes that the data observed represents the actual preference of a user, introducing selection bias [28].

The increasing focus on causal RecSys [30] has led to various methods, each grounded in distinct assumptions aiming to mitigate possible biases. One algorithm named CausE [2] expands on the principles of MF while striving to remove confounding variables by incorporating two training datasets, one large and one small. The first includes insights gathered from a specific chosen recommender policy, while the second derives from an unbiased algorithm (e.g., random). Utilizing the multi-stage learning method results in isolating the environment’s influence on user recommendations.

An alternative method called CausalNeighborhood [26] presents different assumptions and employs the concept of k-nearest neighbors (kNN) to assess the similarity of interactions between users and items. The generated ranked lists are subsequently assessed regarding their causal effect, leading to the formulation of final recommendations.

For the previously stated methods to function, the exposure data is essential. A potential approach for its estimation is CJBPR [37]. This impartial method focuses on understanding user-item relevance and likelihood. Nonetheless, its training necessitates the complete data gathering that includes details regarding the provision of recommendations.

2.2 Selected approaches

As stated in Subsection (2.2), the primary challenge is supplying algorithms with exposure data and its estimation. Therefore, this paper emphasizes the PropCare method [18]. Its primary benefit is that it calculates the likelihood using only user-item interactions. Moreover, similar to CJBPR, it enables the estimation of an item’s relevance probability for a user. This data will be utilized for the alterations outlined in Section (3) along with the rationale and outcomes discussed in Section (4).

Taking into account the causal method for generating recommendations, the DLCE [27] algorithm was chosen. In contrast to CausE, which needs two datasets and the correlation influence found in CausalNeighborhood, it further refines the concepts of MF by directly computing the causal effect using the estimation propositions from [10, 23, 27]. Furthermore, the evaluation of DLCE’s performance against the previously mentioned methods is conducted as a distinct section of the group’s project.

2.3 Related works

Utilizing the selected approaches mentioned in Subsection (2.2), the relevant literature was examined to confirm the importance and distinctiveness of the established objective to investigate the relevance factor of generated recommendations.

Regarding the approaches associated with PropCare, multiple can be addressed. One of the articles presents a method called "CausalX" [16] focused on utilizing LLM for enhancing transparency and reducing bias in the recommendations. Although the integration of explanations ensures causal consistency, this approach does not directly determine the exposure data and demands significant computational resources. A different method outlined in [11] presents a multifactorial bias adjustment using alternating gradient descent. Dealing with more intricate interactions requires explicit ratings data, often lacking in numerous real-world situations. The most recent research mentioned in [1] focuses on stochastic session modeling to represent temporal dynamics. Nonetheless, its limited focus results in reliance on time durations and restricted scalability.

The compilation of associated studies for DLCE is broader and can be categorized into several explored areas. The initial aspect relates to causal inference methods [19, 33, 34], incorporating different research on causal graph modeling [35] to identify intricate relationships or to investigate more reliable estimators [24]. The rise in robustness comes with heightened complexity in execution and calculations, along with a stronger dependence on data specificity. The aim of

alternative debiasing is addressed by [12, 25] or resolved with neural networks usage [7], both of which may lead to overfitting. Research on unlearning biased user preferences [8, 36] or enhancing algorithmic fairness in recommendations [4, 6] has limited causal scope. Additional studies focusing on particular niches (e.g. news [20], cloud services [5]), employing knowledge distillation methods [3], and modeling counterfactual situations [32] are susceptible to reliance on contextual data, diminished causal clarity, and instabilities in training. All the referenced studies emphasize the DLCE advantage as a foundational universal framework for exploring uplift-specific challenges.

2.4 Dataset preprocessing approaches

In order to perform both causal and standard analysis of selected and derived models performance the used datasets needed preprocessing to produce the exact causal effect scores. For this reason two approaches were utilised. The first meta-learner named t-learner described in [15] allows to estimate the uplift by predicting and then subtracting outcomes from treated (with recommendations) and control (without recommendations) groups. The idea is to increase the flexibility by applying separate regression to both groups and then combining them to explain the whole dataset. The second approach from [13] is called DR-learner and its main goal is to construct a pseudo-outcome by applying a two-stage procedure. Firstly, the propensity and outcomes of both control and treatment groups are estimated and then simulate the doubly robust outcome prediction of which allows to get the estimated treatment effect. The motivation behind why the datasets were preprocessed with two different meta-algorithms is presented in Section (4).

3 Problem statement

3.1 Basic notations

For the purpose of intuitive explanation provision alongside the mathematical formulation of the selected approaches described in Subsection (2.2), Table 3.1 contains basic notations that will be used to provide further interpretations.

Referring to the introduced notations, the first definition that is necessary to be presented is how the causal effect is formulated and measured. It is defined as [23]

$$\tau_{u,i} = Y_{u,i}^T - Y_{u,i}^C. \quad (1)$$

Because of the binary nature of results $\tau \in \{-1, 0, 1\}$. The trivial scenario is represented by $\tau = 0$,

Notation	Definition
$u \in \{1, \dots, U\}$	specific user from a set of users
X_u	feature vector of user u
$i \in \{1, \dots, I\}$	specific item from a set of items
X_i	feature vector of item i
pop_i	popularity of item i
X_F	feature vector of frequency interactions (users-items)
$X_F(u), X_F(i)$	feature vector of frequency interactions (user-items or users-item)
$Z_{u,i} \in \{0, 1\}$	indication whether user u was recommended with item i / belongingness to the treatment group
$p_{u,i} \in [0, 1]$	probability that user u was recommended with item i / propensity
$Y_{u,i}^T \in \{0, 1\}$	outcome from the interaction of user u with item i under treatment group
$Y_{u,i}^C \in \{0, 1\}$	outcome from the interaction of user u with item i under control group
$Y_{u,i}$	outcome (rating, purchase, etc.) from the interaction of user u with item i ($Z_{u,i}Y_{u,i}^T + (1 - Z_{u,i})Y_{u,i}^C$)
$R_{u,i} \in \{0, 1\}$	indication whether item i is relevant for user u
$r_{u,i} \in [0, 1]$	probability that item i is relevant for user u

Table 3.1: Basic notations used in the paper

indicating that whether or not there is a recommendation, the item is unappealing to the user. For a positive effect ($\tau = 1$), the model’s recommendation increased the user’s interest in the item, an outcome that wouldn’t have occurred otherwise. The contrary scenario of $\tau = -1$ implies that the suggestion hindered the emergence of user-item interactions. However, the counterfactual nature of causal effect prevents from computing it directly. For this reason the inverse propensity score (IPS) estimator [10, 27] is introduced defined as

$$\hat{\tau}_{u,i} = \frac{Z_{u,i}Y_{u,i}}{p_{u,i}} - \frac{(1 - Z_{u,i})Y_{u,i}}{(1 - p_{u,i})}, \quad (2)$$

where the first and second term are weighted to balance the distribution of underrepresented treatment assignments thus eliminating the bias introduced.

Another important relation [21] to be formalised and used in the definition for PropCare method is defined by the following equation

$$y_{u,i} = p_{u,i}r_{u,i}, \quad (3)$$

where $y_{u,i} = P(Y_{u,i} = 1)$ is the probability of interaction between the user u and item i and $r_{u,i} = P(R_{u,i} = 1)$ is the probability that item i is relevant for user u .

3.2 PropCare

As the current study is the extension of the bachelor’s thesis, the current section will present already modified version of PropCare loss function that proved to provide more reliable propensity and relevance estimates [22]. For the convenience of step-by-step explanation let’s formalize the mathematical problem for the PropCare algorithm first. Its final loss function is constructed with several separate ones. The first component is defined as

$$\mathcal{L}_{naive} = -Y_{u,i} \log(f_p(x_{u,i}; \theta_p) f_r(x_{u,i}; \theta_r)) - (1 - Y_{u,i}) \log(1 - f_p(x_{u,i}; \theta_p) f_r(x_{u,i}; \theta_r)), \quad (4)$$

where $x_{u,i} = f_e(X_u, X_i; \theta_e)$ is the user-item embedding; f_* is trainable MLP with corresponding vector of weights θ_* for either user, propensity or relevance embedding. The intuition behind this term is to reflect the binary cross-entropy relation with the eq. (3). However, the multiplication of propensity and relevance respective output embeddings prevents from differentiation of their individual effects. Thus, the validated approach of popularity introduction [18] helps to formulate the second term defined as following

$$\mathcal{L}_{pop} = -\kappa_{u,i,j} \log(\sigma(\text{sgn}_{i,j} \cdot (f_p(x_{u,i}) - f_p(x_{u,j}))) + \sigma(\text{sgn}_{i,j} \cdot (f_r(x_{u,i}) - f_r(x_{u,j})))), \quad (5)$$

where $\kappa_{u,i,j} = e^{\psi(y_{u,i} - y_{u,j})}$ for hyperparameter $\psi < 0$ is a weighting function, σ is the sigmoid activation function, $\text{sgn}_{i,j} \in \{-1, 1\}$ is the sign of $(pop_i - pop_j)$. The two functions are used to produce the final form of PropCare loss defined as

$$\mathcal{L}_{PropCare} = \sum_{u,i,j} (\mathcal{L}_{naive} + \lambda \mathcal{L}_{pop}) + \mu KL(Q || Beta(\alpha, \beta)) + \lambda_2 \mathcal{L}_{prop} + \lambda_3 \mathcal{L}_{rel} + \nu KL(W || Beta(\alpha, \beta)), \quad (6)$$

where $KL(\cdot || \cdot)$ is the Kullback-Leibler divergence between the distribution of estimated $\hat{p}_{u,i}$ retrieved from Q and the beta distribution with hyperparameters α and β ; λ and μ are hyperparameters balancing the influence of popularity and regularization; $\mathcal{L}_{prop}$, $\mathcal{L}_{rel}$ are binary cross-entropy functions for greater separation of influence derived for more precise estimation of propensity and relevance respectively; $KL(\cdot || \cdot)$ is the Kullback-Leibler divergence between the distribution of estimated $\hat{r}_{u,i}$ retrieved from W and the beta distribution with hyperparameters α and β ; $\lambda_2, \lambda_3, \nu$ are hyperparameters controlling the influence of different terms. The loss function is minimized to obtain the estimated propensities $\hat{p}_{u,i}$ and predict $\hat{Z}_{u,i}$ through $Norm(\hat{p}_{u,i}) \geq \epsilon$ with ϵ being a threshold parameter and $Norm$ a Z -score normalization function.

3.3 DLCE

After the propensities are trained and exposure data is estimated it is passed to the DLCE algorithm to produce recommendations. For each pair of item i and user u the score is estimated defined as

$$\hat{s}_{u,i} = f_M(X_u, X_i; \theta_M), \quad (7)$$

where $f_M(\cdot)$ is a trainable MLP. Items are then sorted by $\hat{s}_{u,i}$ for each user and ranked producing a vector $r(\hat{s}_{u,i})$. From eq. (7) the average metric function is derived presented as

$$R(M) = \frac{1}{UI} \sum_u \sum_i \lambda(r(\hat{s}_{u,i})) \tau_{u,i}, \quad (8)$$

where $\lambda(r(\hat{s}_{u,i})) = \lambda_{u,i} = -r(\hat{s}_{u,i})$ is a weighting function and M denotes model. From it the empirical risk minimization (ERM) function becomes $\operatorname{argmin}_M(-R(M))$ containing the counterfactual parameter τ . By substituting it with IPS estimation from eq. (2) and introducing capping for propensity deviation constraint [27], the capped IPS version is derived as

$$R^{CIPS}(M) = \lambda_{u,i} \left(\frac{Z_{u,i} Y_{u,i}}{\max(p_{u,i}, \chi^T)} - \frac{(1 - Z_{u,i}) Y_{u,i}}{\max((1 - p_{u,i}), \chi^C)} \right) = \lambda_{u,i} \tilde{\tau}_{u,i}, \quad (9)$$

where χ^T, χ^C are capping parameters bounding the denominators.

The final step is to estimate the weighting function $\lambda_{u,i}$, otherwise called as causal average rank (CAR). To perform it the local loss is introduced as

$$\delta_{u,i} = -\lambda_{u,i} \tilde{\tau}_{u,i} = r(\hat{s}_{u,i}) \tilde{\tau}_{u,i}, \quad (10)$$

where $r(\hat{s}_{u,i}) = (1 + \sum_{j \neq i} \mathbf{1}(\hat{s}_{u,j} > \hat{s}_{u,i}))$. By introducing the approximation of the non-differentiable indicator function [17] and applying some transformational changes the final loss is describes as

$$\begin{aligned} \delta_{u,i} &= r(\hat{s}_{u,i}) \tilde{\tau}_{u,i} \\ &= r(\hat{s}_{u,i}) \tilde{Y}_{u,i}^T - (I - r(\hat{s}_{u,i})) \tilde{Y}_{u,i}^C - I \tilde{Y}_{u,i}^C \\ &\leq \left(\sum_{j \neq i} \log(1 + \exp(-\omega \hat{s}_{u,i,j})) \right) \frac{Z_{u,i} Y_{u,i}}{\max(p_{u,i}, \chi^T)} - \\ &\quad - \left(\sum_{j \neq i} \log(1 + \exp(\omega \hat{s}_{u,i,j})) \right) \frac{(1 - Z_{u,i}) Y_{u,i}}{\max((1 - p_{u,i}), \chi^C)} \\ &= \mathcal{L}_{DLCE}, \end{aligned} \quad (11)$$

where $I = 1 + \sum_{j \neq i} \mathbf{1}(\hat{s}_{u,j} > \hat{s}_{u,i}) + \sum_{j \neq i} \mathbf{1}(\hat{s}_{u,i} > \hat{s}_{u,j})$; $\hat{s}_{u,i,j} = \hat{s}_{u,i} - \hat{s}_{u,j}$; ω is a hyperparameter

for approximation. The obtained loss function is then used to produce the ranking and provide recommendations for the user based on it.

In this research a new version of DLCE was investigated. Main intuition for it was the competitive performance of ranking by relevance probabilities which can be seen in Section (4.2). It is formalised as following:

$$\hat{s}_{u,i}^{Rel} = f_M(X_u, X_i + \iota * r_{u,:}; \theta_M), \quad (12)$$

where ι is the weight for the estimated relevance probability of an user-item pair. The weight parameter was selected by checking parameters from the set $\{1, 2, 5, 10, 20, 50\}$. $\iota = 10$ as it uniformly provided with better results and balanced by restraining DLCE to reflect relevance ranking a lot. The research also tried other combinations of relevance estimates addition ($f_M(X_u, X_i; \theta_M) + \iota * r_{u,:}$ and $f_M(X_u, X_i + \iota_2 * r_{u,:}; \theta_M) + \iota_2 * r_{u,:}$) but they did not provide enhanced performance compared to initially introduced variant.

The two other types of the DLCE presented below were derived from [22]. During experiments with DunnHumby both of them proved to enhance the performance of the used model. The first one is the altered loss function represented by the eq.(11) which was formalised as

$$\begin{aligned} \mathcal{L}_{DLCE}^{Mod} &= \hat{r}(\hat{s}_{u,i}) \tilde{\tau}_{u,i}' \\ &= \hat{r}(\hat{s}_{u,i}) (\tilde{\tau}_{u,i} + \phi \tilde{\tau}_{u,i}^{rel}) \\ &= \hat{r}(\hat{s}_{u,i}) (\tilde{\tau}_{u,i} + \phi (\frac{R_{u,i} Y_{u,i}}{\max(r_{u,i}, \chi^T)} - \frac{(1 - R_{u,i}) Y_{u,i}}{\max((1 - r_{u,i}), \chi^C)})), \end{aligned} \quad (13)$$

where $\tilde{\tau}_{u,i}^{rel}$ measures ITE of item i relevance on the user u and ϕ regulates the effect it has on the total loss. The second one incorporates the frequency of user-item positive interactions based on history. Three different variants of such transformations are represented as

$$\begin{aligned} \hat{s}_{u,i}^F &= f_M(X_u, X_i; \theta_M) + X_F, \\ \hat{s}_{u,i}^{FU} &= f_M(X_u + X_F(u), X_i; \theta_M), \\ \hat{s}_{u,i}^{FI} &= f_M(X_u, X_i + X_F(i); \theta_M) \end{aligned} \quad (14)$$

4 Experiment setup and results

4.1 Setup description

For the evaluation of PropCare and DLCE T-ECD dataset were used. T-ECD is a large-scale synthetic cross-domain dataset for recommender systems research, created by T-Bank's RecSys R&D team. It captures real-world e-commerce interaction patterns across multiple domains while ensuring complete anonymity through synthetic generation. Thus, in contrast to

DunnHumby used in [22], the used dataset can be described as "less" synthetic as it doesn't necessarily produce every item and every user interaction. The current research focuses on the small version of the dataset for computational constraints reasons. For uplift specific task datasets from two domains are used: Marketplace and Retail. Marketplace contains information about browsing and interacting with items in an e-commerce marketplace. Retail - interactions within a retail delivery service, including cart additions and completed orders. These datasets have explicit identification of whether or not the item was recommended to the user and what was the final output.

Nevertheless, the datasets still needed to be preprocessed as there was no causal effect ($\tau_{u,i}$) for later evaluation. As a result, preprocessing of Marketplace was done using t-learner [15] and preprocessing of Retail domain with DR-learner [13]. For the first one firstly the propensity scores were predicted using the treated column derived from the dataset (a user is recommended is *subdomain* column shows "u2i" or "i2i"). Then two RandomForestRegressor models are used to separately train on control and treatment groups and predict the causal effect by subtracting the predicted outcomes. For the Retail dataset the preprocessing consists of firstly estimating propensities through 3-fold cross-fitting (only "i2i" treatment for this domain) and then predicting the causal effects by deploying a GradientBoostingRegressor DR pseudo-outcomes. For both datasets outcome is considered to be marked as "1" if the user made any kind of interaction other than "view". The predicted column is then transformed to $\{-1, 0, 1\}$ by analysis. The intuition behind two different approaches for datasets preprocessing I=lies in the fact that the initial choice was to use t-learner for both datasets. But during its application to Retail domain it tended to provide with suspiciously high number of positive causal effects estimates (which was not the case for Marketplace). Thus, another approach was decided to be taken and DR estimation resulted in satisfiable performance.

The basic principles of the training that comprises both exposure data estimation and recommendation generation is described by the Algorithm 1. The base realisation of DLCE was taken from the files that are attached to [27]. The base realisation of PropCare algorithm was taken from open GitHub repository¹ from the authors of the original paper [18]. In addition, some of the hyperparameters were changed. The code is stored at the GitHub repository².

From [22] the same two sets of metrics were selected for evaluation (left ones are for standard performance, right ones for causal):

¹<https://github.com/mediumboat/PropCare/tree/main>

²<https://github.com/RomanGorelsky/Uplift-Modeling-RecSys>

$$\begin{aligned}
NDCG@k &= \frac{DCG@k}{IDCG@k} = \frac{\sum_u \sum_{i=1}^{k_{rank}} \frac{Y_{u,i}}{\log_2 i + 1}}{\sum_u \sum_{i=1}^{k_{ideal}} \frac{Y_{u,i}}{\log_2 i + 1}} & CP@k &= \frac{1}{U} \sum_u \sum_i \frac{\mathbf{1}(\text{rank}_u(\hat{s}_{u,i}) \leq k) \tau_{u,i}}{k} \\
Recall@k &= \frac{\sum_u \sum_{i=1}^k Y_{u,i}}{\sum_u \sum_i Y_{u,i}} & CDCG &= \frac{1}{U} \sum_u \sum_i \frac{\tau_{u,i}}{\log_2(1 + \text{rank}_u(\hat{s}_{u,i}))} \\
Precision@k &= \frac{\sum_u \sum_{i=1}^k Y_{u,i}}{k}
\end{aligned}$$

In addition, two new metrics called Sparsity-Adjusted Debiased Incremented Response and Rank-Constrained Average Uplift defined by:

$$\begin{aligned}
SA-DIR@K &= \frac{1}{|U_{\geq m}|} \sum_{u \in U_{\geq m}} \frac{1}{K} \sum_{i \in \text{top}_K(u)} \left[\frac{Z_{u,i} Y_{u,i}}{p_{u,i}} - \frac{(1 - Z_{u,i}) Y_{u,i}}{1 - p_{u,i}} \right] \\
RCAU@K &= \frac{\sum_{u \in U} \frac{\min(K, n_u)}{K^2} \sum_{i \in \text{top}_{\min(K, n_u)}(u)} \tau_{u,i}}{\sum_{u \in U} \left(\frac{\min(K, n_u)}{K} \right)^2},
\end{aligned}$$

SA-DIR@k shows the average debiased causal effect (through DR estimation) for users that have at least m number of interactions ($U_{\geq m}$, $m = 3$ for both settings). RCAU@k weights the causal effect for users based on the number of interactions they have in the dataset (the less there are, the greater the penalty). These weighted-related metrics were introduced to reveal new sides about models performance to help differentiate them. The described metrics mean and standard deviation were collected across 3 consecutive runs.

In conclusion, the metric comparison between Algorithm 1 and its modifications was performed using the recommendation ranking derived from the relevance probabilities generated by PropCare (row "Relevance (R)"), along with two baseline rankings based on popularity and a modified variant called personal popularity calculated from the training dataset (rows "Popularity (P)" and "Personal Popular (PP)").

4.2 Results description

The current section presents and analyses the results received from the application of the selected approaches to the chosen datasets. The best metrics for corresponding setups in each column can be distinguished by their bold highlighting (firstly the largest mean is prioritised, then the lowest standard deviation).

4.2.1 Marketplace domain

Before the discussion of the received results for the current domain an important notion must be made. The Marketplace dataset is highly sparse in terms of repeated user-item interactions. Thus, the resulted differences in metrics between different versions of DLCE are almost negligible (to the range of $[1e-3, 1e-4]$). In addition, the distribution of estimated causal effects is highly concentrated around 0. Nevertheless, the conducted analysis still helps to retrieve useful information about applied approaches performance.

Firstly, the results with base and frequency-dependent DLCE versions were received and they described in the Table 4.1.

Table 4.1: Metrics on $T - ECD(\text{Marketplace})$ (Mean $\pm$ Std. Error)

Method	NDCG@10	Recall@10	Precision@10
Relevance (R)	0.2264 ± 0.0079	0.3401 ± 0.0084	0.0135 ± 0.0005
Popularity (P)	0.2570 ± 0.0000	0.3672 ± 0.0000	0.0152 ± 0.0000
Personal Popular (PP)	0.2579 ± 0.0000	0.3674 ± 0.0000	0.0153 ± 0.0000
DLCE ($\hat{s}_{u,i}$)	0.2215 ± 0.0011	0.3359 ± 0.0021	0.0132 ± 0.0001
DLCE ($\hat{s}_{u,i}^F$)	0.2243 ± 0.0011	0.3358 ± 0.0021	0.0136 ± 0.0001
DLCE ($\hat{s}_{u,i}^{FI}$)	0.2230 ± 0.0010	0.3363 ± 0.0020	0.0134 ± 0.0001
DLCE ($\hat{s}_{u,i}^{FU}$)	0.2233 ± 0.0011	0.3359 ± 0.0020	0.0135 ± 0.0001

Method	CP@10	CP@100	CDCG
Relevance (R)	0.0074 ± 0.0050	0.0376 ± 0.0029	0.2834 ± 0.0149
Popularity (P)	-0.0114 ± 0.0000	0.0210 ± 0.0000	0.2378 ± 0.0000
Personal Popular (PP)	-0.0115 ± 0.0000	0.0210 ± 0.0000	0.2371 ± 0.0000
DLCE ($\hat{s}_{u,i}$)	0.0075 ± 0.0002	0.0379 ± 0.0002	0.2828 ± 0.0002
DLCE ($\hat{s}_{u,i}^F$)	0.0064 ± 0.0002	0.0374 ± 0.0001	0.2806 ± 0.0002
DLCE ($\hat{s}_{u,i}^{FI}$)	0.0070 ± 0.0002	0.0378 ± 0.0002	0.2823 ± 0.0001
DLCE ($\hat{s}_{u,i}^{FU}$)	0.0073 ± 0.0002	0.0379 ± 0.0002	0.2828 ± 0.0002

Method	RCAU@10	SA-DIR@10
Relevance (R)	0.0027 ± 0.0064	0.0619 ± 0.0697
Popularity (P)	-0.0212 ± 0.0000	0.1217 ± 0.1400
Personal Popular (PP)	-0.0214 ± 0.0000	0.1216 ± 0.1429
DLCE ($\hat{s}_{u,i}$)	0.0029 ± 0.0003	0.1403 ± 0.1552
DLCE ($\hat{s}_{u,i}^F$)	0.0015 ± 0.0003	0.1379 ± 0.1526
DLCE ($\hat{s}_{u,i}^{FI}$)	0.0023 ± 0.0003	0.1392 ± 0.1539
DLCE ($\hat{s}_{u,i}^{FU}$)	0.0026 ± 0.0003	0.1376 ± 0.1545

Method	Coverage for SA-DIR@10
All Methods	0.6499

The received metrics give several interesting insights. Considering the different metrics, the non-causal approaches tend to provide better performance on standard types whereas the DLCE variants excel on causal ones. In addition, the frequency-based DLCE approach helps to increase the performance on the standard metrics. All this aligns with [22].

However, unlike the conclusions made in the bachelor thesis there are new interesting findings. Firstly, the causal metrics reveal that frequency incorporation in a setup of high user-item interaction sparsity leads to a worsening effect. Also, the Personal Popular approach (which

was highly competitive in DunnHumby setting [22] sometimes leading to better performance than DLCE) presents worse performance with even negative values for CP@10. Finally, an important notion must be made about relevance approach that shows competitive performance with DLCE but struggles from high standard deviation.

Taking into consideration the newly introduced causal metrics, their analysis also unravels interesting patterns. The RCAU@10 metric (as weighted version of CP@10) shows that performance of all approaches is lower. However, for Popularity and Personal Popular the magnitude is a lot higher than for DLCE. Analysis of DA-DIR@10 metric reveal that for this setup it is quite unstable (with coverage around 65% overall). If regarded in terms of mean maximisation, the standard DLCE excels whereas for std minimisation relevance also shows acceptable range (it falls below zero less than other ones).

Following the ideas of [22] the next step was to perform tuning with $\mathcal{L}_{DLCE}^{Mod}$ and perform experiments with best parameter of ϕ and selected threshold for relevance. All the simulations results can be found by the repository link on GitHub (for convenience of analysis they were transformed to a table with highlighted values for each metric). As a result, for the Marketplace domain $\phi = 0.1$ and threshold equal to 0.6 were selected. The obtained results are presented on the Table 4.2.

Table 4.2: Metrics with $\mathcal{L}_{DLCE}^{Mod}$ on $T - ECD(Marketplace)$ (Mean $\pm$ Std. Error)

Method	NDCG@10	Recall@10	Precision@10
Relevance (R)	0.2264 $\pm$ 0.0079	0.3401 $\pm$ 0.0084	0.0135 $\pm$ 0.0005
Popularity (P)	0.2570 $\pm$ 0.0000	0.3672 $\pm$ 0.0000	0.0152 $\pm$ 0.0000
Personal Popular (PP)	0.2579 $\pm$ 0.0000	0.3674 $\pm$ 0.0000	0.0153 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}$)	0.2217 $\pm$ 0.0006	0.3357 $\pm$ 0.0004	0.0133 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}^F$)	0.2243 $\pm$ 0.0006	0.3352 $\pm$ 0.0006	0.0136 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}^{FI}$)	0.2229 $\pm$ 0.0006	0.3358 $\pm$ 0.0005	0.0134 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}^{FU}$)	0.2226 $\pm$ 0.0005	0.3356 $\pm$ 0.0007	0.0134 $\pm$ 0.0000

Method	CP@10	CP@100	CDCG
Relevance (R)	0.0074 $\pm$ 0.0050	0.0376 $\pm$ 0.0029	0.2834 $\pm$ 0.0149
Popularity (P)	-0.0114 $\pm$ 0.0000	0.0210 $\pm$ 0.0000	0.2378 $\pm$ 0.0000
Personal Popular (PP)	-0.0115 $\pm$ 0.0000	0.0210 $\pm$ 0.0000	0.2371 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}$)	0.0073 $\pm$ 0.0001	0.0376 $\pm$ 0.0001	0.2834 $\pm$ 0.0003
DLCE ($\hat{s}_{u,i}^F$)	0.0062 $\pm$ 0.0002	0.0371 $\pm$ 0.0001	0.2805 $\pm$ 0.0004
DLCE ($\hat{s}_{u,i}^{FI}$)	0.0068 $\pm$ 0.0002	0.0376 $\pm$ 0.0001	0.2822 $\pm$ 0.0006
DLCE ($\hat{s}_{u,i}^{FU}$)	0.0066 $\pm$ 0.0002	0.0375 $\pm$ 0.0001	0.2819 $\pm$ 0.0004

Method	RCAU@10	SA-DIR@10
Relevance (R)	0.0027 $\pm$ 0.0064	0.0619 $\pm$ 0.0697
Popularity (P)	-0.0212 $\pm$ 0.0000	0.1217 $\pm$ 0.1400
Personal Popular (PP)	-0.0214 $\pm$ 0.0000	0.1216 $\pm$ 0.1429
DLCE ($\hat{s}_{u,i}$)	0.0026 $\pm$ 0.0002	0.1403 $\pm$ 0.1569
DLCE ($\hat{s}_{u,i}^F$)	0.0012 $\pm$ 0.0002	0.1381 $\pm$ 0.1538
DLCE ($\hat{s}_{u,i}^{FI}$)	0.0019 $\pm$ 0.0002	0.1397 $\pm$ 0.1554
DLCE ($\hat{s}_{u,i}^{FU}$)	0.0017 $\pm$ 0.0002	0.1410 $\pm$ 0.1570

Unlike conclusions made in [22] in the current setup $\mathcal{L}_{DLCE}^{Mod}$ doesn't help to elevate the performance. On the contrary the model shows slightly worse results compared to Table 4.1 overall (except for SA-DIR@10 where a slight increase persists).

Following the failure of previously developed relevance-based method to excel, the research dived into discovering new approaches. For this reason and for the reason of competitive performance of relevance ranking on both Tables 4.1 and 4.2 the $\hat{s}_{u,i}^{Rel}$ approach was developed. Its results are presented in the Table 4.3.

Table 4.3: Metrics with DLCE $\hat{s}_{u,i}^{Rel}$ version on $T - ECD(Marketplace)$ (Mean $\pm$ Std. Error)

Method	NDCG@10	Recall@10	Precision@10
Relevance (R)	0.2264 $\pm$ 0.0079	0.3401 $\pm$ 0.0084	0.0135 $\pm$ 0.0005
Popularity (P)	0.2570 $\pm$ 0.0000	0.3672 $\pm$ 0.0000	0.0152 $\pm$ 0.0000
Personal Popular (PP)	0.2579 $\pm$ 0.0000	0.3674 $\pm$ 0.0000	0.0153 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}^{Rel}$)	0.2228 $\pm$ 0.0015	0.3364 $\pm$ 0.0021	0.0134 $\pm$ 0.0002
DLCE ($\hat{s}_{u,i}^{F-Rel}$)	0.2248 $\pm$ 0.0010	0.3366 $\pm$ 0.0022	0.0136 $\pm$ 0.0001
DLCE ($\hat{s}_{u,i}^{FI-Rel}$)	0.2238 $\pm$ 0.0011	0.3367 $\pm$ 0.0020	0.0135 $\pm$ 0.0001
DLCE ($\hat{s}_{u,i}^{FU-Rel}$)	0.2254 $\pm$ 0.0012	0.3363 $\pm$ 0.0021	0.0137 $\pm$ 0.0001

Method	CP@10	CP@100	CDCG
Relevance (R)	0.0074 $\pm$ 0.0050	0.0376 $\pm$ 0.0029	0.2834 $\pm$ 0.0149
Popularity (P)	-0.0114 $\pm$ 0.0000	0.0210 $\pm$ 0.0000	0.2378 $\pm$ 0.0000
Personal Popular (PP)	-0.0115 $\pm$ 0.0000	0.0210 $\pm$ 0.0000	0.2371 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}^{Rel}$)	0.0080 $\pm$ 0.0017	0.0379 $\pm$ 0.0003	0.2846 $\pm$ 0.0042
DLCE ($\hat{s}_{u,i}^{F-Rel}$)	0.0072 $\pm$ 0.0017	0.0375 $\pm$ 0.0002	0.2826 $\pm$ 0.0041
DLCE ($\hat{s}_{u,i}^{FI-Rel}$)	0.0076 $\pm$ 0.0016	0.0378 $\pm$ 0.0002	0.2838 $\pm$ 0.0039
DLCE ($\hat{s}_{u,i}^{FU-Rel}$)	0.0068 $\pm$ 0.0016	0.0374 $\pm$ 0.0002	0.2816 $\pm$ 0.0038

Method	RCAU@10	SA-DIR@10
Relevance (R)	0.0027 $\pm$ 0.0064	0.0619 $\pm$ 0.0697
Popularity (P)	-0.0212 $\pm$ 0.0000	0.1217 $\pm$ 0.1400
Personal Popular (PP)	-0.0214 $\pm$ 0.0000	0.1216 $\pm$ 0.1429
DLCE ($\hat{s}_{u,i}^{Rel}$)	0.0035 $\pm$ 0.0022	0.1568 $\pm$ 0.1519
DLCE ($\hat{s}_{u,i}^{F-Rel}$)	0.0025 $\pm$ 0.0021	0.1562 $\pm$ 0.1535
DLCE ($\hat{s}_{u,i}^{FI-Rel}$)	0.0030 $\pm$ 0.0021	0.1561 $\pm$ 0.1528
DLCE ($\hat{s}_{u,i}^{FU-Rel}$)	0.0020 $\pm$ 0.0020	0.1503 $\pm$ 0.1489

The results presented push to two conclusions. On the one hand, the developed approach helped to increase the performance of all DLCE approaches across almost all metrics. On the other, the relevance estimates also introduced the noise and std increased. In the Marketplace setup the developed approach can be considered as success as even though the range of metrics increased the context suggests it to be negligible compared to increase in the mean. However, $\hat{s}_{u,i}^{Rel}$ prediction needs to be studied on the Retail domain.

4.2.2 Retail domain

Unlike the Marketplace setting, Retail domain is much less sparse in terms user-item interaction history. This is related to its nature as of goods-related actions (buying groceries, etc.).

The experiments for Retail domain follow Marketplace structure with initial results received represented by Table 4.4.

Table 4.4: Metrics on $T - ECD(Retail)$ (Mean $\pm$ Std. Error)

Method	NDCG@10	Recall@10	Precision@10
Relevance (R)	0.7065 $\pm$ 0.0177	0.3092 $\pm$ 0.0015	0.6208 $\pm$ 0.0126
Popularity (P)	0.7542 $\pm$ 0.0000	0.3156 $\pm$ 0.0000	0.6590 $\pm$ 0.0000
Personal Popular (PP)	0.7402 $\pm$ 0.0000	0.3155 $\pm$ 0.0000	0.6500 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}$)	0.6356 $\pm$ 0.0025	0.3039 $\pm$ 0.0007	0.5619 $\pm$ 0.0019
DLCE ($\hat{s}_{u,i}^F$)	0.7015 $\pm$ 0.0013	0.3089 $\pm$ 0.0005	0.6209 $\pm$ 0.0013
DLCE ($\hat{s}_{u,i}^{FI}$)	0.6635 $\pm$ 0.0023	0.3058 $\pm$ 0.0006	0.5869 $\pm$ 0.0021
DLCE ($\hat{s}_{u,i}^{FU}$)	0.6686 $\pm$ 0.0070	0.3056 $\pm$ 0.0010	0.5916 $\pm$ 0.0058

Method	CP@10	CP@100	CDCG
Relevance (R)	0.5527 $\pm$ 0.0176	0.3934 $\pm$ 0.0046	10.1643 $\pm$ 0.0850
Popularity (P)	0.5768 $\pm$ 0.0000	0.3977 $\pm$ 0.0000	10.2653 $\pm$ 0.0000
Personal Popular (PP)	0.5372 $\pm$ 0.0000	0.3909 $\pm$ 0.0000	10.1357 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}$)	0.4770 $\pm$ 0.0008	0.3502 $\pm$ 0.0004	9.6196 $\pm$ 0.0061
DLCE ($\hat{s}_{u,i}^F$)	0.5112 $\pm$ 0.0024	0.3846 $\pm$ 0.0012	10.0095 $\pm$ 0.0136
DLCE ($\hat{s}_{u,i}^{FI}$)	0.4895 $\pm$ 0.0030	0.3664 $\pm$ 0.0027	9.8097 $\pm$ 0.0280
DLCE ($\hat{s}_{u,i}^{FU}$)	0.4806 $\pm$ 0.0099	0.3528 $\pm$ 0.0035	9.6435 $\pm$ 0.0465

Method	RCAU@10	SA-DIR@10
Relevance (R)	0.5463 $\pm$ 0.0183	-2.4129 $\pm$ 0.7716
Popularity (P)	0.5713 $\pm$ 0.0000	-2.8104 $\pm$ 0.8427
Personal Popular (PP)	0.5302 $\pm$ 0.0000	-2.4295 $\pm$ 0.8298
DLCE ($\hat{s}_{u,i}$)	0.4679 $\pm$ 0.0008	-1.3348 $\pm$ 0.7225
DLCE ($\hat{s}_{u,i}^F$)	0.5033 $\pm$ 0.0025	-2.0581 $\pm$ 0.8138
DLCE ($\hat{s}_{u,i}^{FI}$)	0.4809 $\pm$ 0.0031	-1.6224 $\pm$ 0.7523
DLCE ($\hat{s}_{u,i}^{FU}$)	0.4717 $\pm$ 0.0103	-1.2894 $\pm$ 0.6966

Method	Coverage for SA-DIR@10
All Methods	0.9060

The results reveal a different picture compared to previous domain. For the Retail setup the Popularity ranking (followed closely by Personal Popular) excels in all previous standard and causal metrics and DLCE falls behind other approaches. Nevertheless, this time performance of frequency-based approaches is better with DLCE ($\hat{s}_{u,i}^F$) providing the best among other variants. This highly resembles the situation with $DH_{personalised}$ setting from [22]. However, unlike the bachelor’s thesis results relevance ranking shows competitive performance and is even better than DLCE (but its std is the largest). This conclusion is a supporting factor to verify $\hat{s}_{u,i}^{Rel}$ in this setting.

But the most interesting part considers the new introduced causal metrics, their analysis also unravels interesting patterns. RCAU@10 once again down weights a bit the causal effect of all models. The interesting part is the analysis of SA-DIR@10. Firstly, for Retail it has a lot higher coverage (a bit higher than 90%). Secondly, unlike the previous causal metrics it ranks Popularity and Personal Popular results as the 1st and 2nd worst. Also, the best performing DLCE ($\hat{s}_{u,i}^F$) turns out to be the worst one among other modifications in this regard. Finally, the

best approaches are DLCE ($\hat{s}_{u,i}$) and ($\hat{s}_{u,i}^{FU}$) with a big advantage (even best case Popularity SA-DIR@10 cannot beat worst ($\hat{s}_{u,i}^{FU}$)). This analysis is quite informative in the sense that previously it was considered that DLCE ($\hat{s}_{u,i}^F$) excels over other approaches. But now the focus is changed towards DLCE ($\hat{s}_{u,i}^{FU}$).

Following the structure, tuning was performed and $\phi = 0.3$ with threshold 0.7 were selected (once again all the simulations results are presented in the GitHub repository). The obtained results are presented on the Table 4.5.

Table 4.5: Metrics with $\mathcal{L}_{DLCE}^{Mod}$ on $T - ECD(Retail)$ (Mean $\pm$ Std. Error)

Method	NDCG@10	Recall@10	Precision@10
Relevance (R)	0.7065 $\pm$ 0.0177	0.3092 $\pm$ 0.0015	0.6208 $\pm$ 0.0126
Popularity (P)	0.7542 $\pm$ 0.0000	0.3156 $\pm$ 0.0000	0.6590 $\pm$ 0.0000
Personal Popular (PP)	0.7402 $\pm$ 0.0000	0.3155 $\pm$ 0.0000	0.6500 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}$)	0.6363 $\pm$ 0.0007	0.3042 $\pm$ 0.0001	0.5629 $\pm$ 0.0009
DLCE ($\hat{s}_{u,i}^F$)	0.7004 $\pm$ 0.0008	0.3091 $\pm$ 0.0001	0.6201 $\pm$ 0.0005
DLCE ($\hat{s}_{u,i}^{FI}$)	0.6635 $\pm$ 0.0009	0.3059 $\pm$ 0.0001	0.5872 $\pm$ 0.0009
DLCE ($\hat{s}_{u,i}^{FU}$)	0.6651 $\pm$ 0.0054	0.3055 $\pm$ 0.0004	0.5886 $\pm$ 0.0047

Method	CP@10	CP@100	CDCG
Relevance (R)	0.5527 $\pm$ 0.0176	0.3934 $\pm$ 0.0046	10.1643 $\pm$ 0.0850
Popularity (P)	0.5768 $\pm$ 0.0000	0.3977 $\pm$ 0.0000	10.2653 $\pm$ 0.0000
Personal Popular (PP)	0.5372 $\pm$ 0.0000	0.3909 $\pm$ 0.0000	10.1357 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}$)	0.4753 $\pm$ 0.0001	0.3491 $\pm$ 0.0006	9.6121 $\pm$ 0.0017
DLCE ($\hat{s}_{u,i}^F$)	0.5076 $\pm$ 0.0011	0.3832 $\pm$ 0.0004	9.9913 $\pm$ 0.0065
DLCE ($\hat{s}_{u,i}^{FI}$)	0.4866 $\pm$ 0.0004	0.3643 $\pm$ 0.0006	9.7824 $\pm$ 0.0045
DLCE ($\hat{s}_{u,i}^{FU}$)	0.4745 $\pm$ 0.0087	0.3507 $\pm$ 0.0034	9.6183 $\pm$ 0.0477

Method	RCAU@10	SA-DIR@10
Relevance (R)	0.5463 $\pm$ 0.0183	-2.4129 $\pm$ 0.7716
Popularity (P)	0.5713 $\pm$ 0.0000	-2.8104 $\pm$ 0.8427
Personal Popular (PP)	0.5302 $\pm$ 0.0000	-2.4295 $\pm$ 0.8298
DLCE ($\hat{s}_{u,i}$)	0.4662 $\pm$ 0.0001	-1.3137 $\pm$ 0.7102
DLCE ($\hat{s}_{u,i}^F$)	0.4996 $\pm$ 0.0011	-2.0174 $\pm$ 0.7588
DLCE ($\hat{s}_{u,i}^{FI}$)	0.4778 $\pm$ 0.0005	-1.6027 $\pm$ 0.7259
DLCE ($\hat{s}_{u,i}^{FU}$)	0.4653 $\pm$ 0.0090	-1.2458 $\pm$ 0.7244

Following similar conclusions for Table 4.2, $\mathcal{L}_{DLCE}^{Mod}$ was unable to enhance the performance except for SA-DIR@10. This concludes that at least for less synthetic situations $\mathcal{L}_{DLCE}^{Mod}$ is not the best choice.

Finally, the simulations were performed for DLCE $\hat{s}_{u,i}^{Rel}$ and presented in Table 4.6.

Once again DLCE $\hat{s}_{u,i}^{Rel}$ proves to be a potential enhancement of standard approach. Even though the std increased once again metrics increased by greater magnitude compared to previous domain. This allowed to make DLCE ($\hat{s}_{u,i}^{FU-Rel}$) a more competitive approach. For example, it now outperforms relevance ranking in all standard metrics and in CP@100 or personal popular in all previously introduced causal metrics and RAUC@10.

However, SA-DIR@10 once again gives interesting insights. First of all, every DLCE approach outperforms other methods. However, DLCE ($\hat{s}_{u,i}^{FU-Rel}$) is now the worst among all (given

Table 4.6: Metrics with DLCE $\hat{s}_{u,i}^{Rel}$ version on $T - ECD(Retail)$ (Mean $\pm$ Std. Error)

Method	NDCG@10	Recall@10	Precision@10
Relevance (R)	0.7065 $\pm$ 0.0177	0.3092 $\pm$ 0.0015	0.6208 $\pm$ 0.0126
Popularity (P)	0.7542 $\pm$ 0.0000	0.3156 $\pm$ 0.0000	0.6590 $\pm$ 0.0000
Personal Popular (PP)	0.7402 $\pm$ 0.0000	0.3155 $\pm$ 0.0000	0.6500 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}$)	0.6354 $\pm$ 0.0027	0.3039 $\pm$ 0.0002	0.5621 $\pm$ 0.0021
DLCE ($\hat{s}_{u,i}^F$)	0.6938 $\pm$ 0.0029	0.3082 $\pm$ 0.0008	0.6145 $\pm$ 0.0028
DLCE ($\hat{s}_{u,i}^{FI}$)	0.6578 $\pm$ 0.0029	0.3057 $\pm$ 0.0005	0.5821 $\pm$ 0.0030
DLCE ($\hat{s}_{u,i}^{FU}$)	0.7177 $\pm$ 0.0042	0.3103 $\pm$ 0.0007	0.6354 $\pm$ 0.0035

Method	CP@10	CP@100	CDCG
Relevance (R)	0.5527 $\pm$ 0.0176	0.3934 $\pm$ 0.0046	10.1643 $\pm$ 0.0850
Popularity (P)	0.5768 $\pm$ 0.0000	0.3977 $\pm$ 0.0000	10.2653 $\pm$ 0.0000
Personal Popular (PP)	0.5372 $\pm$ 0.0000	0.3909 $\pm$ 0.0000	10.1357 $\pm$ 0.0000
DLCE ($\hat{s}_{u,i}^{Rel}$)	0.4784 $\pm$ 0.0027	0.3523 $\pm$ 0.0019	9.6423 $\pm$ 0.0218
DLCE ($\hat{s}_{u,i}^{F-Rel}$)	0.5060 $\pm$ 0.0029	0.3815 $\pm$ 0.0032	9.9744 $\pm$ 0.0271
DLCE ($\hat{s}_{u,i}^{FI-Rel}$)	0.4868 $\pm$ 0.0030	0.3619 $\pm$ 0.0029	9.7517 $\pm$ 0.0377
DLCE ($\hat{s}_{u,i}^{FU-Rel}$)	0.5403 $\pm$ 0.0068	0.3965 $\pm$ 0.0020	10.1479 $\pm$ 0.0295

Method	RCAU@10	SA-DIR@10
Relevance (R)	0.5463 $\pm$ 0.0183	-2.4129 $\pm$ 0.7716
Popularity (P)	0.5713 $\pm$ 0.0000	-2.8104 $\pm$ 0.8427
Personal Popular (PP)	0.5302 $\pm$ 0.0000	-2.4295 $\pm$ 0.8298
DLCE ($\hat{s}_{u,i}$)	0.4694 $\pm$ 0.0028	-1.3077 $\pm$ 0.6845
DLCE ($\hat{s}_{u,i}^F$)	0.4979 $\pm$ 0.0030	-1.9044 $\pm$ 0.6671
DLCE ($\hat{s}_{u,i}^{FI}$)	0.4780 $\pm$ 0.0031	-1.5506 $\pm$ 0.6396
DLCE ($\hat{s}_{u,i}^{FU}$)	0.5335 $\pm$ 0.0071	-2.3853 $\pm$ 0.7725

that it improved in previous causal metrics). This reveals an interesting tradeoff where the addition of relevance and increase in previously introduced causal performance leads to a decreased average debiased causal effect overall.

5 Conclusions and future prospects

The current paper extends the research on the topic of uplift modeling started in the bachelor’s thesis [22]. Conducted research helps to reveal new insights which could be a potential work for the author’s master thesis.

Overall, the perfumed analysis helps to discover the following: (i) in the setting of limited (sparse) user-item interaction information the addition of frequency to DLCE model’s final prediction leads to worsening of metrics, (ii) standard approaches like Popularity and Personal Popular finally show their inability for stable performance across different domains in a less synthesised setting (Marketplace vs Retail) which leads to conclusion that DLCE and Relevance rankings should only be prioritised, (iii) newly introduced causal metrics (especially SA-DIR@K) reveal interesting trade-off in relevance addition.

Considering additional future prospects of the work, several ideas are interesting for the research. Firstly, the assumption about the availability of user-item frequency interaction data

reveals its high influence on final predictions results. Thus, the cold start is a real problem for study. Secondly, it would be beneficial to look at what were the values of newly introduced causal metrics for scenarios from the author's bachelor's thesis. Finally, the research should focus on more intense comparison of DLCE ($\hat{s}_{u,i}^F$) and ($\hat{s}_{u,i}^{FU}$) and try to derive the uniform solution.

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A Appendix

Algorithm 1 PropCare and DLCE Training

```

1: Input: Observed training data  $D$ ,  $\eta$ ,  $\gamma$ ,  $\omega$ ,  $\chi^T$ ,  $\chi^C$ 
2: Output: Weights DLCE ( $\theta_{DLCE}$ )
3:
4: Initialize PropCare weights:  $\theta_{PropCare}$ 
5: for  $e = 1$  to  $E$  do
6:   for pair  $(u, i)$  in  $D$  do
7:     Compute  $\mathcal{L}_{naive}$  from eq. (4)
8:     Sample an item  $j$  for each pair from  $I \setminus \{i\}$ 
9:     Compute  $\mathcal{L}_{pop}$  from eq. (5)
10:    Compute  $\mathcal{L}_{PropCare}$  from eq. (??)
11:  end for
12:  Update  $\theta_{PropCare}$  with backpropagation
13: end for
14: Obtain exposure data from  $\theta_{PropCare}$ 
15: Initialize DLCE weights:  $\theta_{DLCE}$ 
16: for  $n = 1$  to  $N$  do
17:   Sample a pair  $(u, i)$  from  $D$ 
18:   Sample an item  $j$  for each pair from  $I \setminus \{i\}$ 
19:   if  $Z_{u,i} = 1$  then
20:      $\theta_{DLCE} = \theta_{DLCE} + \eta \left( -\frac{1}{\max(p_{u,i}, \chi^T)} \frac{\partial}{\partial t} (\log(1 + \exp(\omega \hat{s}_{u,i,j}))) - \gamma \theta_{DLCE} \right)$ 
21:   else
22:      $\theta_{DLCE} = \theta_{DLCE} + \eta \left( -\frac{1}{\max((1 - p_{u,i}), \chi^C)} \frac{\partial}{\partial t} (\log(1 + \exp(-\omega \hat{s}_{u,i,j}))) - \gamma \theta_{DLCE} \right)$ 
23:   end if
24: end for
25:
26: Return:  $\theta_{DLCE}$ 

```
